Little Rooms: How Cells Were Discovered

Learning area: Science — Biological sciences; Science as a Human Endeavour

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Worksheet

Part A

1. Nouns: *scientist, microscope* · Verb: *adjusted* · Adjectives: *patient, brass* (*brass* is a noun used as an adjective/classifier — accept and discuss) · Adverb: *carefully*. (*The* is a determiner/article.)
2. *There are many different types of cells in the human body; each has its own job.* (Errors: *their*→*There* (capital + spelling), *type's*→*types*, *cell's*→*cells*, *it's*→*its*, and the comma splice → semicolon/full stop/"and each".) Comma splice counts as the fifth error.
3. *Hooke published a book in 1665, which contained detailed drawings of insects.* (Or: *In 1665 Hooke published a book which contained…*)

Part C

1. The compartments reminded him of small rooms (Latin *cella*). He was actually seeing the walls of dead plant cells; the living contents had gone.
2. Order: Hooke (1) → van Leeuwenhoek (2) → Schleiden (3) → Schwann (4) → Virchow (5) → electron microscope (6).
3. 1. All living things are made of one or more cells. (2) The cell is the basic unit of structure and function. (3) All cells come from cells that already existed.
4. Stated: nobody else could see what he claimed (his lenses were better than anyone else's). Inferred: he had no university training / was a tradesman, not a recognised scientist; he wrote in Dutch from abroad; the claims were extraordinary (invisible living things everywhere). Reward reasoning about status and evidence.
5. Technology limited observation (blurry, coloured images until better microscopes in the 1830s); observations had to be gathered across many organisms before a general rule could be made; a wrong idea (spontaneous formation of cells) had to be disproved by observing cell division.
6. Either view, if justified. Strong answers note Virchow publicised and popularised the idea and connected it to medicine, but Remak supplied key evidence and wasn't fully acknowledged — so credit is at least partly unfair. Look for "although/however" balance.
7. Schleiden believed cells formed spontaneously from fluid; the third idea says all cells come from pre-existing cells, which directly contradicts him.

Connections

8. The **cell wall** — it is tough, non-living material that remains after the cell dies; the membrane, cytoplasm, nucleus and other living contents break down and dry out.
9. **Cell wall** (support/rigidity — no skeleton), **chloroplasts** (photosynthesis — plants make their own food), **large permanent vacuole** (stores water, keeps the cell firm). Animals eat food and have skeletons/other support.

10. All cells come from pre-existing cells (bacteria divide) — and they are made of a cell / are the basic unit of life. It shows cell theory applies even to organisms without a nucleus, and explains how infections spread (bacteria multiply by division, not spontaneously).
11. a. 1 cm = 10 mm; 10 ÷ 0.02 = **500 cells** b) 0.02 × 200 = **4 mm** c) Through Hooke's ~50× microscope a 0.02 mm cell would appear only 1 mm wide — too small to see detail; at 200× it appears 4 mm, large enough to see. Bacteria are far smaller than human cells, so only the stronger lens could reveal them. *(Hooke's compound microscope magnification is commonly estimated at ~30–50×.)*

Part D

12.

a. [*When he examined a thin slice of cork*], **he saw** [*that it was full of tiny, empty compartments*]. — *When* (and *that*). Main clause: *he saw*.
b. **Not complex** — it is **compound**: two main clauses joined by the coordinating conjunction *but*. Good discussion point.
c. **Some members doubted him**, [*since nobody else could see* [*what he claimed to see*]]. — *since* (plus embedded *what* clause).
d. **Virchow is usually given the credit**, [*although historians now point out* [*that he relied heavily on Remak's evidence*]]. — *although*.

13. Models:

a. *After Schwann compared animal and plant tissue, he concluded that animals are made of cells.*
b. *Scientists could not see fine details inside cells because early microscopes blurred images.*
c. *Van Leeuwenhoek, who was a cloth merchant, discovered bacteria.*

14. *Because early microscopes blurred images, scientists could not see…* — fronting the cause emphasises the **reason** (technology) and creates suspense; end position emphasises the **result**. Comma rule must be applied.
15. Model (26 words): *Self-taught Dutch lens-maker Antonie van Leeuwenhoek used powerful microscopes to discover single-celled organisms, including bacteria; doubted at first, his findings were accepted once Hooke confirmed them.*
16. Model (34 words): *Although Hooke named cells in 1665, cell theory developed over two centuries as better microscopes allowed scientists such as van Leeuwenhoek, Schleiden, Schwann, Remak and Virchow to gather evidence, showing that science keeps changing.*

Part E

17. a. small drawings / drawings of small things b) the study of cells c) green (pigment) of leaves d) the study of small living things e) a small animal f) white (blood) cell
18. Any plausible invention with a definition, e.g. *chloroscope* — an instrument for examining green pigments.

Part F — marking notes

A strong paragraph:

- **T:** e.g. *The history of the cell shows that scientific knowledge grows and changes whenever new tools allow scientists to collect better evidence.*
- **Evidence:** Hooke's 1665 microscope revealing cork "cells"; van Leeuwenhoek's superior lenses (>200×) revealing bacteria; better microscopes in the 1830s enabling Schleiden (1838) and Schwann (1839); observation of cell division (Remak, Virchow 1855) overturning Schleiden's idea; electron microscope (1930s) revealing mitochondria.
- **Explanation:** links each tool to new evidence and changed ideas — not just a list of events.
- **Link:** returns to the prompt with a judgement (e.g. *knowledge is never finished*).

Criterion	Developing	Strong
Argument	retells history	uses history as evidence for a claim
Evidence	vague	specific names + dates, ≥2
Sentences	mostly simple/compound	≥3 controlled complex sentences, correct commas
Vocabulary	general	precise, includes roots (*microscope, microbiology, cytology*)

Part G

19. Strong answers: viruses fail idea 1 (not made of cells) and cannot reproduce alone (idea 3 says cells come from cells; viruses rely on cells), so by cell theory they are not alive; *Although* they contain genetic material, reproduce and evolve inside hosts, which are features of life… Reward a reasoned conclusion either way.
20. *Schleiden believed that cells could form spontaneously from the fluid inside organisms…* followed by *The answer came from careful observation of cells dividing.* — signalled by "There was, however, a gap" and the verb *believed* (vs. evidence-based *observed*). Also accept Hooke's misunderstanding (empty rooms vs living cells).

Homework

1. Differentiation: the process by which cells change their shape and contents to become specialised for one job as an embryo develops.
2.

Cell	Feature	How it helps
Red blood cell	biconcave disc; no nucleus	large surface area to absorb oxygen; more room for haemoglobin
Neuron	long, thin extensions; many branches	carries signals over long distances; connects with thousands of other cells
Root hair cell	long finger-like extension	larger surface area to absorb water from soil
Guard cell	curved, in pairs	open/close stomata to control water loss and CO_2 entry

3. Advantage: more space for haemoglobin → carries more oxygen. Disadvantage: cannot repair itself → lives only ~4 months.

4. e.g. *Although every cell in your body began from a single fertilised egg, your cells are not all the same. / Because they lack a nucleus, however, they cannot repair themselves… / …which gives them a large surface area…*
5. Model (20 words): *As embryos develop, cells become specialised, so each type's structure — like red blood cells or root hairs — suits its function.*
6. haematology — study of blood; neurology — study of nerves/nervous system; stomata — "mouths" (pores) in leaves; erythrocyte — red (blood) cell.
7. Any genuine response.

Teacher notes

- **Common misconceptions:** treating compound sentences (*and/but/so*) as complex; missing comma after a fronted subordinate clause; summaries that copy phrases or keep examples.
- **If the student finishes early:** have them rewrite ¶4 for a Year 4 audience — discuss what they simplified and why (register).
- **If the student struggles:** colour-code clauses with highlighters; do Q9 orally first.
- **Sources / fact-check (verified common history-of-science facts):**
  - Hooke, *Micrographia* (1665); coined "cell" for cork compartments.
  - Van Leeuwenhoek: Delft draper; single-lens microscopes, magnifications estimated up to ~270×; letters to the Royal Society from 1673, including the 1676 letter describing "animalcules"; observations confirmed by Hooke in 1677.
  - Schleiden 1838 (plants), Schwann 1839 (animals).
  - Remak observed animal cell division (1850s); Virchow popularised *omnis cellula e cellula* (1855).
  - Electron microscope: first built in 1931 (Ruska & Knoll), developed through the 1930s.
  - Human body: roughly 30–37 trillion cells (estimates vary) → text says "tens of trillions".
  - Red blood cell lifespan ~120 days; longest human neurons (sciatic pathway) ~1 m.
- **Next week builds on:** complex sentences → passive voice and nominalisation in *From Bill to Act* (civics).

Fast-finisher bank — answers & marking notes

F1. The list shows the huge range of living things — large/small, plant/animal/fungus, familiar/personal — to stress that cells are *universal*; ending with "your own hand" makes it personal. The dashes set off the list as an interruption/illustration, adding emphasis.

F2. *usually given the credit*; *historians now point out*; *almost like crystals*; *occasionally forced*; *often disagreed*; *sometimes completely wrong*. Historical claims are interpretations, evidence can be incomplete, and credit is disputed — cautious language makes the writing more accurate and credible.

F3. a) Full of holes passing *through* it. b) Capitalised common noun *Honey-comb*; hyphenated compound; adverb *exceedingly plainly*; phrase *but that the pores of it were not regular* (archaic construction). c) The primary source gives Hooke's own words, what he saw and how he described it at the time; the article gives context, hindsight (what he was *really* seeing), and the later significance.

F4. Check: correct subordinator for each type; comma after each fronted clause; embedded clause bracketed by **two** commas (e.g. *The microscope, which Hooke improved, revealed…*).

F5. Model (≈50 words): *For a long time, microscopes were not very good. Then, about 190 years ago, two German scientists looked closely at lots of living things. Schleiden found that all plants are made of cells. Schwann found that animals are too. So they decided every living thing is made of cells.* Changes: shorter sentences, removed job titles/technical words (*botanist, physiologist, tissues*), used concrete time reference, dropped detail about blurring.

F6. Scale makes the **long gap** (1680s → 1838) and the clustering of discoveries in the 1830s–1850s visible — it shows progress was uneven and depended on technology.

F7. Look for: first-person, past/present tense control, period-appropriate voice (no modern slang), accurate details, required sentence types, emotional complexity (wonder + anxiety about the Royal Society).

F8. *For:* clear rule, rewards sharing findings, publication allows checking (peer review). *Against:* ignores those who supplied evidence (Remak), favours people with access to journals or fame, discoveries are often collaborative/simultaneous (e.g. Darwin and Wallace, Newton and Leibniz).

F9. Check facts against the student's named source; reward linking the discovery to the technology (X-ray crystallography → DNA structure; electron microscopy → organelles; cryo-EM → proteins).